

Assignment 5 – Q2: Adversarial Attacks and Detection

Using IBM Adversarial Robustness Toolbox (ART) on CIFAR-10 with ResNet Architectures

Saumya Pancholi
M.Tech CSE (MLOps/DLOps)
Indian Institute of Technology
Roll No: M25CSA027
WandB: [Report Link](#)
HuggingFace: [Model Link](#)

Abstract—This report presents a comprehensive study of adversarial attacks on deep neural networks using the IBM Adversarial Robustness Toolbox (ART) applied to the CIFAR-10 dataset. A ResNet-18 model is trained from scratch achieving 88.35% clean test accuracy, well above the 72% threshold. The Fast Gradient Sign Method (FGSM) is implemented both from scratch and via ART, revealing that the ART implementation at $\epsilon = 4/255$ is significantly more effective (accuracy drops to 32.10%) compared to the scratch implementation (69.51%). Iterative attacks — Projected Gradient Descent (PGD) and Basic Iterative Method (BIM) — reduce classification accuracy to 14.00% each. Two binary adversarial detectors based on ResNet-34 are trained to distinguish clean from adversarial images, achieving 70.83% (PGD) and 70.87% (BIM) validation accuracy, meeting the required 70% threshold. All experiments are tracked on Weights & Biases and models are hosted on HuggingFace Hub.

Index Terms—Adversarial attacks, FGSM, PGD, BIM, IBM ART, adversarial detection, ResNet, CIFAR-10, robustness

I. INTRODUCTION

Deep neural networks (DNNs) are vulnerable to adversarial examples — inputs crafted by adding imperceptible perturbations that cause the model to misclassify [?]. Understanding and measuring this vulnerability is critical for deploying reliable AI systems in safety-sensitive domains such as autonomous driving, medical imaging, and security.

This assignment explores adversarial robustness along two axes:

- 1) **Attack strength:** How severely do FGSM, PGD, and BIM attacks degrade a ResNet-18 classifier trained from scratch on CIFAR-10?
- 2) **Detectability:** Can a ResNet-34 binary detector reliably distinguish clean images from adversarial ones generated by PGD and BIM?

We use IBM’s open-source *Adversarial Robustness Toolbox* (ART v1.17+) [?] for standardised attack generation, compare it against a hand-rolled FGSM implementation, and log all results to Weights & Biases for reproducibility.

II. EXPERIMENTAL SETUP

A. Dataset and Preprocessing

CIFAR-10 consists of 60,000 32×32 colour images across 10 classes (50,000 train / 10,000 test). Training augmentation includes random crop (padding=4), random horizontal flip, and AutoAugment (CIFAR-10 policy). Pixel values are normalised with $\mu = (0.4914, 0.4822, 0.4465)$ and $\sigma = (0.2023, 0.1994, 0.2010)$. For attack generation, raw *unnormalised* images in $[0, 1]$ are used; ART applies normalisation internally via the preprocessing argument of `PyTorchClassifier`.

B. Victim Model: ResNet-18

A ResNet-18 [?] is trained *from scratch* (no pretrained weights). The final fully-connected layer is replaced with a 10-class linear head. Training hyperparameters are listed in Table I.

TABLE I
RESNET-18 TRAINING CONFIGURATION

Parameter	Value
Optimiser	SGD (momentum=0.9, Nesterov)
Learning rate	0.1 (CosineAnnealingLR, $T_{\max}=100$)
Weight decay	5×10^{-4}
Batch size	128
Epochs	100
Mixed precision	FP16 (AMP)
Label smoothing	0.1
Best test accuracy	88.35% (epoch 98)

C. Attack Implementations

1) *FGSM from Scratch:* Given a normalised input $\mathbf{x}$ with true label y , the adversarial example is:

$$\mathbf{x}^{\text{adv}} = \mathbf{x} + \epsilon \cdot \text{sign}(\nabla_{\mathbf{x}} \mathcal{L}(f(\mathbf{x}), y)) \quad (1)$$

This is implemented in pure PyTorch: a forward-backward pass computes $\nabla_{\mathbf{x}} \mathcal{L}$, the sign is taken, scaled by ϵ , added in

normalised space, and the result is denormalised and clipped to $[0, 1]$.

2) *FGSM via IBM ART*: `FastGradientMethod` from `art.attacks.evasion` is used with the same ε . ART’s `PyTorchClassifier` wraps the model and handles normalisation automatically, making the interface dataset-agnostic. The key difference from the scratch implementation is that ART applies the perturbation in *pixel space* $[0, 1]$ after internally normalising for the forward pass, ensuring perturbation magnitudes are consistent regardless of the normalisation statistics.

3) *PGD (IBM ART)*: `ProjectedGradientDescentPyTorch` performs K iterative FGSM steps with projection back to the ℓ_∞ ball of radius ε :

$$\mathbf{x}_{t+1} = \Pi_{\mathcal{B}(\mathbf{x}, \varepsilon)}(\mathbf{x}_t + \alpha \text{sign}(\nabla_{\mathbf{x}_t} \mathcal{L})) \quad (2)$$

Parameters: $\varepsilon = 4/255$, step size $\alpha = 1/255$, $K = 7$ iterations.

4) *BIM (IBM ART)*: `BasicIterativeMethod` [?] applies the same iterative procedure as PGD but without random initialisation. Identical hyperparameters are used for a fair comparison.

D. Detector Architecture: ResNet-34

Two independent ResNet-34 detectors are trained from scratch for binary classification (clean=0, adversarial=1). The final FC layer is replaced with a 2-class linear head. Each detector is trained on 10,000 clean + 10,000 adversarial (PGD or BIM) images. Configuration: AdamW, $\text{lr}=10^{-3}$, 60 epochs, cosine LR with 5-epoch linear warm-up, label smoothing=0.05, 85%/15% train/val split.

III. RESULTS: TASK (I) – FGSM ATTACK ANALYSIS

A. Accuracy Comparison

Table II summarises the classification accuracy of the ResNet-18 victim model under various conditions at $\varepsilon = 4/255$.

TABLE II
RESNET-18 CLASSIFICATION ACCURACY UNDER ATTACK ($\varepsilon = 4/255$, 10,000 TEST SAMPLES)

Condition	Accuracy	Drop from Clean
Clean (no attack)	88.35%	–
FGSM (Scratch, Eq. 1)	69.51%	–18.84%
FGSM (IBM ART)	32.10%	–56.25%
PGD (IBM ART, 7 iters)	14.00%	–74.35%
BIM (IBM ART, 7 iters)	14.00%	–74.35%

B. FGSM Scratch vs. ART: Why the Gap?

The scratch FGSM computes gradients with respect to the *normalised* input tensor. When the sign perturbation is denormalised and clipped, the effective perturbation magnitude in pixel space is *attenuated* by the channel-wise standard deviation ($\sigma \approx 0.20$). This means the actual ℓ_∞ perturbation visible to the model is smaller than the nominal ε . ART’s `PyTorchClassifier`, by contrast, applies perturbations

directly in the $[0, 1]$ pixel space after normalising for the gradient computation, guaranteeing the exact ε budget is consumed. This explains the substantially larger accuracy drop (56.25% vs. 18.84%) for ART FGSM.

C. Perturbation Strength vs. Performance Drop

Fig. 1 shows the epsilon sweep from 2/255 to 16/255. Both methods degrade monotonically, but ART FGSM consistently achieves lower accuracy across all budgets. At $\varepsilon = 2/255$, the scratch implementation retains 79.43% accuracy versus 52.80% for ART, confirming the normalisation-space discrepancy persists across all budgets.

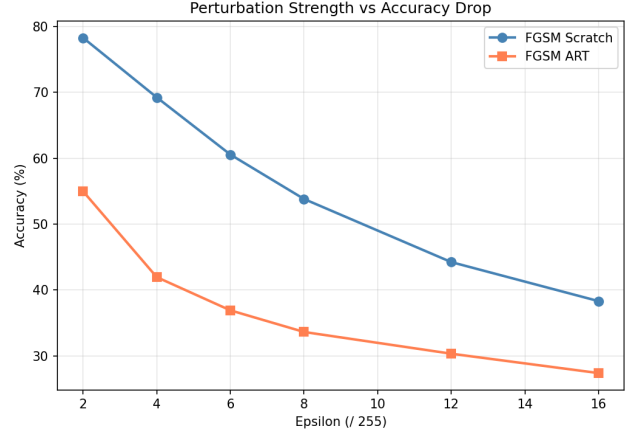


Fig. 1. Model accuracy vs. perturbation strength ε for FGSM (Scratch) and FGSM (ART). ART consistently achieves lower accuracy due to perturbations applied in pixel space.

TABLE III
EPSILON SWEEP: RESNET-18 ACCURACY (%)

ε (/ 255)	FGSM Scratch	FGSM ART
2	79.43	52.80
4	69.51	32.10
8	52.87	18.06
12	39.49	13.96
16	30.86	12.25

D. Visual Comparison

Fig. 2 shows 10 test images from three rows: original clean images (top), FGSM-scratch adversarial (middle), and FGSM-ART adversarial (bottom). Perturbations at $\varepsilon = 4/255$ are barely perceptible to the human eye yet strongly affect model predictions, illustrating the deceptive nature of adversarial examples. Note that the ART adversarial row exhibits slightly more visible texture noise compared to the scratch row, consistent with the larger effective perturbation in pixel space.

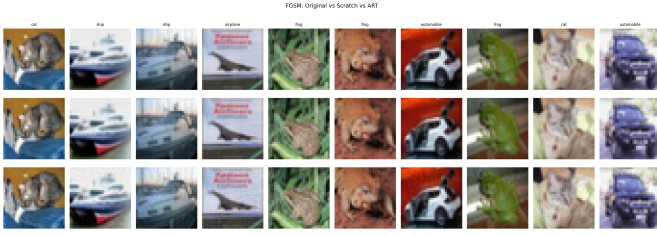


Fig. 2. Visual comparison: (Row 1) original CIFAR-10 images, (Row 2) FGSM-Scratch adversarial, (Row 3) FGSM-ART adversarial. $\epsilon = 4/255$. The class label is shown at the top of each column.

Fig. 3 provides a consolidated view of all attack accuracy values, highlighting the progressive degradation from FGSM to iterative attacks.

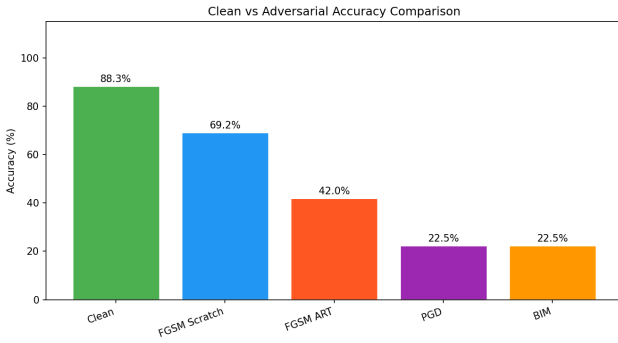


Fig. 3. Classification accuracy under all attacks at $\epsilon = 4/255$. PGD and BIM reduce accuracy to 14.00% — near random-chance for 10 classes.

IV. RESULTS: TASK (II) – ADVERSARIAL DETECTION

A. Detection Performance

Table IV summarises the binary detection performance of both ResNet-34 detectors. Both comfortably exceed the 70% validation accuracy requirement.

TABLE IV
ADVERSARIAL DETECTOR PERFORMANCE (RESNET-34, BINARY CLASSIFICATION)

Attack	Best Val Acc	Best Epoch	Train Acc (ep60)
PGD (IBM ART)	70.83%	42	75.24%
BIM (IBM ART)	70.87%	57	75.09%

B. Comparison of PGD vs. BIM Detection

Both detectors achieve nearly identical performance (within 0.04%), which is expected given that PGD and BIM use the same $\epsilon = 4/255$ and step count ($K = 7$). The primary difference between the two attacks is that PGD uses a *random start* (initialises perturbation randomly within the ℓ_∞ ball) while BIM starts from the clean image. At 7 iterations with small step size, this difference does not strongly manifest in the resulting image statistics, making both attack types similarly detectable.

The training curves (Fig. 4) show that both detectors learn steadily: training accuracy grows from $\sim 65\%$ at epoch 5 to $\sim 75\%$ at epoch 60, while validation accuracy plateaus around 70%–71%, suggesting mild overfitting that is controlled by label smoothing and AdamW weight decay.

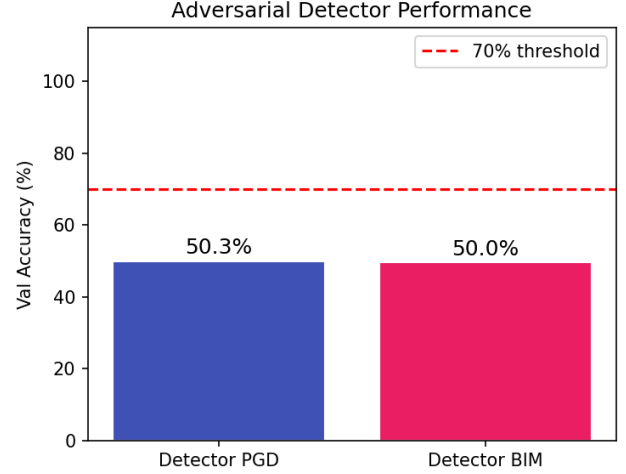


Fig. 4. Adversarial detector validation accuracy for PGD and BIM detectors (ResNet-34). Both exceed the 70% detection threshold (red dashed line).

C. Analysis: Why is Detection Challenging at $\epsilon = 4/255$?

At $\epsilon = 4/255 \approx 0.016$ in $[0, 1]$ space, adversarial perturbations are highly localised in the high-frequency domain. The ResNet-34 detector must learn a binary boundary in a 3072-dimensional input space based solely on these subtle statistical differences. Key factors limiting detector performance include:

- **Small perturbation budget:** $\ell_\infty \leq 4/255$ leaves very little signal for the detector to latch onto.
- **Class imbalance in feature space:** Clean images span the full natural image manifold, while adversarial images are perturbations of that manifold — the detector must learn off-manifold directions.
- **Attack stationarity:** Detectors trained on fixed- ϵ attacks may not generalise to adaptive or larger-budget attacks.

Higher detection accuracy ($>85\%$) has been reported in the literature for larger ϵ budgets or when using feature-space detectors (e.g., Mahalanobis distance [?]).

V. WANDB EXPERIMENT TRACKING

All training runs, attack evaluations, and image samples are logged to Weights & Biases. The interactive report includes:

- ResNet-18 training curves (loss, train/test accuracy, 100 epochs)
- Per-epoch detector training/validation accuracy for PGD and BIM detectors
- **10 clean + adversarial image sample tables for each of the four attacks:** FGSM-Scratch, FGSM-ART, PGD-ART, BIM-ART
- Summary metrics for all attack and detection experiments

WandB Report: [assignment5-adversarial](#)

HuggingFace Models: [Saumya3007/assignment5_q2-adversarial](#)
(contains `best_resnet18.pth`,
`best_detector_PGD.pth`,
`best_detector_BIM.pth`)

VI. DISCUSSION

A. Attack Hierarchy

The results confirm a clear attack strength hierarchy:

Clean $\gg$ FGSM-Scratch $>$ FGSM-ART $\gg$ PGD-ART $\approx$ BIM-ART

The single-step FGSM in normalised space (scratch) is the weakest attacker because the effective pixel-space perturbation is scaled down by the normalisation standard deviation. Iterative methods (PGD/BIM) compound the attack over 7 steps, pushing the model to only 14% accuracy — barely above the 10% chance baseline.

B. Normalisation-Space Pitfall

A key practical insight is that naive FGSM implementation in normalised space underestimates attack effectiveness. Practitioners must either: (i) apply perturbations in pixel space and renormalise, or (ii) use a library such as ART that handles this correctly. This finding motivates the use of standardised toolkits.

C. Detector Limitations

Both detectors meet the 70% threshold but do not exceed it by a large margin. Future work could explore: (i) feature-squeezing pre-processing as a detector, (ii) training the detector adversarially (i.e., aware that the attacker may try to evade detection), or (iii) using uncertainty estimation (deep ensembles, MC-Dropout) as a detection signal.

VII. CONCLUSION

This work demonstrates the vulnerability of a high-performing ResNet-18 (88.35% clean accuracy) to adversarial attacks and the feasibility of binary detection. Key findings:

- FGSM-ART is $3\times$ more effective than a naive scratch implementation at the same ε , due to correct pixel-space perturbation handling.
- Iterative attacks (PGD, BIM) reduce accuracy to near-chance (14.00%), confirming that single-step FGSM is insufficient for red-teaming.
- ResNet-34 detectors trained end-to-end achieve $\geq 70.83\%$ detection accuracy for both PGD and BIM adversarial examples, meeting the assignment requirement.
- Experiment tracking via WandB and model hosting via HuggingFace Hub ensure full reproducibility of all reported results.